Section 1.5 Sequences & summation

数列与求和

[AIME-2001-II-Q03] | Part A (Q1-Q5)

Let $x_1 = 211, x_2 = 375, x_3 = 420, x_4 = 523$ and $x_n = x_{n-1} - x_{n-2} + x_{n-3} - x_{n-4}$ for $n \geq 5$. Find $x_{531} + x_{753} + x_{975}$.

数列初值为 $211, 375, 420, 523$，且 $n \geq 5$ 时 $x_n = x_{n-1} - x_{n-2} + x_{n-3} - x_{n-4}$。求 $x_{531} + x_{753} + x_{975}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 1.5 Sequences & summation

数列与求和

[AIME-2002-I-Q04] | Part A (Q1-Q5)

For $a_k = 1/(k^2+k)$, positive integers $m < n$ satisfy $\sum_{k=m}^{n-1} a_k = 1/29$. Find $m+n$.

令 $a_k = 1/(k^2+k)$，正整数 $m < n$ 满足 $\sum_{k=m}^{n-1} a_k = 1/29$，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2004-I-Q02] | Part A (Q1-Q5)

Set A contains m consecutive integers with sum $2m$. Set B contains $2m$ consecutive integers with sum m. Their largest elements differ in absolute value by 99. Find m.

集合 A 有 m 个连续整数，总和 $2m$；集合 B 有 $2m$ 个连续整数，总和 m。两集合最大元之差绝对值为 99，求 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2005-II-Q03] | Part A (Q1-Q5)

An infinite geometric series sums to 2005. Squaring each term gives a series with sum 20050. If the original ratio is reduced $m/n > 0$, find $m + n$.

无穷等比级数和为 2005，每项平方后的级数和为 20050。若原公比为最简正分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2008-II-Q01] | Part A (Q1-Q5)

Let $N = 100^2 + 99^2 - 98^2 - 97^2 + 96^2 + 95^2 - \cdots + 4^2 + 3^2 - 2^2 - 1^2$. Find $N \bmod 1000$.

令 $N = 100^2 + 99^2 - 98^2 - 97^2 + 96^2 + 95^2 - \cdots + 4^2 + 3^2 - 2^2 - 1^2$，求其除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q03] | Part A (Q1-Q5)

The interior angles of a convex eighteen-gon are distinct increasing integers in arithmetic progression, measured in degrees. Find the least angle.

凸十八边形的内角度数为严格递增的整数等差数列。求最小内角。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q05] | Part A (Q1-Q5)

A geometric series has sums $S_{2011} = 200$ and $S_{4022} = 380$. Find S_{6033}.

某等比数列前 2011 项和为 200，前 4022 项和为 380，求前 6033 项和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2012-I-Q02] | Part A (Q1-Q5)

An arithmetic sequence sums to 715. Adding $2k-1$ to its kth term changes the total to 836. Find the sum of its original first, middle, and last terms.

一等差数列各项和为 715。将第 k 项增加 $2k-1$ 后，总和为 836。求原数列首项、中间项及末项之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2012-II-Q02] | Part A (Q1-Q5)

Two geometric sequences have the same nonzero common ratio, first terms $27, 99$, and $a_{15} = b_{11}$. Find a_9.

两个等比数列有相同的非零公比，首项分别为 $27, 99$，且 $a_{15} = b_{11}$。求 a_9。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2013-II-Q03] | Part A (Q1-Q5)

A 119-cm candle takes $10k$ seconds to burn its kth centimeter from the top. If the full burn lasts T, its remaining height at time $T/2$ is h. Find $10h$.

一根一百一十九厘米的蜡烛，从顶端起第 k 厘米需燃烧 $10k$ 秒。完全燃尽时间为 T，在 $T/2$ 时剩余高度为 h，求 $10h$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2015-I-Q01] | Part A (Q1-Q5)

Let $A = \sum_{k=1}^{19}(2k-1)(2k) + 39$ and $B = 1 + \sum_{k=1}^{19}(2k)(2k+1)$. Calculate $|A - B|$.

令 $A = \sum_{k=1}^{19}(2k-1)(2k) + 39$，$B = 1 + \sum_{k=1}^{19}(2k)(2k+1)$。求 $|A - B|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q01] | Part A (Q1-Q5)

For $|r| < 1$, define $S(r) = 12 \sum_{k=0}^{\infty} r^k$. If $|a| < 1$ and $S(a)S(-a) = 2016$, find $S(a) + S(-a)$.

对 $|r| < 1$，定义 $S(r) = 12 \sum_{k=0}^{\infty} r^k$。若 $|a| < 1$ 且 $S(a)S(-a) = 2016$，求 $S(a) + S(-a)$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q05] | Part A (Q1-Q5)

A reader finishes a 374-page book over at least two days. On day j, she reads $n+j-1$ pages in $t+j-1$ minutes, with positive integers n, t. Her total reading time is 319 minutes. Find $n+t$.

某人用至少两天读完 374 页书。第 j 天读 $n+j-1$ 页，用时 $t+j-1$ 分钟，n, t 为正整数。总用时 319 分钟。求 $n+t$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-II-Q01] | Part A (Q1-Q5)

Three people initially hold increasing peanut counts $A < B < C$, in geometric progression, totaling 444. After they eat respectively $5, 9, 25$ peanuts, the counts in the same order form an arithmetic progression. Find A.

三人起初的花生数 $A < B < C$ 成等比数列，总计 444。他们依次吃掉 $5, 9, 25$ 颗后，按同一顺序的剩余数量成等差数列。求 A。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2023-II-Q01] | Part A (Q1-Q5)

Six tree harvests form an arithmetic progression with sum 990, and the largest harvest is twice the smallest. Find the largest harvest.

六棵树的果实数成等差数列，总数 990，最大项为最小项的两倍。求最大项。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2026-II-Q01] | Part A (Q1-Q5)

Sum the tenth terms of all integer arithmetic sequences whose first term is 4 and which contain both 24 and 34.

求所有首项为 4 且包含 $24, 34$ 的整数等差数列的第十项之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2002-II-Q06] | Part B (Q6-Q10)

Find the nearest integer to $1000\sum_{n=3}^{10000} 1/(n^2-4)$.

求最接近 $1000\sum_{n=3}^{10000} 1/(n^2-4)$ 的整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2003-II-Q08] | Part B (Q6-Q10)

A sequence begins $1440, 1716, 1848$ and is the termwise product of two arithmetic sequences. Find its eighth term.

数列首三项为 $1440, 1716, 1848$，它由两个等差数列对应项相乘得到，求第八项。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2004-II-Q09] | Part B (Q6-Q10)

Positive integers a_j begin with $a_1 = 1$. Every triple at odd starting index is geometric; every triple at even starting index is arithmetic. Given $a_9 + a_{10} = 646$, let a_n be the largest sequence term below 1000. Find $n + a_n$.

正整数数列首项为一，奇数起始下标的每组三项等比，偶数起始下标的每组三项等差。已知 $a_9 + a_{10} = 646$，设 a_n 为小于一千的最大项，求 $n + a_n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2008-I-Q06] | Part B (Q6-Q10)

The top row of a triangular sum array is $1, 3, 5, \ldots, 99$. Each lower cell is the sum of the two just above it, until one bottom cell remains. How many array entries are divisible by 67?

三角求和阵首行为 $1, 3, 5, \ldots, 99$，每下一行少一项，每项为上方相邻两项之和，直至底部一项。求全阵中被六十七整除的项数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.5 Sequences & summation

数列与求和

[AIME-2008-II-Q06] | Part B (Q6-Q10)

Sequences a_n, b_n both obey $u_n = u_{n-1} + u_{n-1}^2/u_{n-2}$ for $n \geq 2$. Initial terms are $(a_0, a_1) = (1, 1)$ and $(b_0, b_1) = (1, 3)$. Find b_{32}/a_{32}.

两数列都满足 $u_n = u_{n-1} + u_{n-1}^2/u_{n-2}$，初值分别为 $(1, 1)$ 与 $(1, 3)$，求 b_{32}/a_{32}。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2009-I-Q07] | Part B (Q6-Q10)

A real sequence has $a_1 = 1$ and $5^{a_{n+1}-a_n} - 1 = 1/(n + 2/3)$. Find the smallest integer $k > 1$ for which a_k is an integer.

实数列满足 $a_1 = 1, 5^{a_{n+1}-a_n} - 1 = 1/(n + 2/3)$，求使 a_k 为整数的最小整数 $k > 1$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2009-I-Q08] | Part B (Q6-Q10)

Take all positive differences between elements of $\{2^0, 2^1, \ldots, 2^{10}\}$. If their sum is N, find $N \bmod 1000$.

取集合 $\{2^0, 2^1, \ldots, 2^{10}\}$ 中元素两两的所有正差，和为 N，求除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2014-II-Q07] | Part B (Q6-Q10)

Let $f(x) = (x^2 + 3x + 2)^{\cos \pi x}$. Find the sum of positive integers n such that $|\sum_{k=1}^{n} \log_{10} f(k)| = 1$.

令 $f(x) = (x^2 + 3x + 2)^{\cos \pi x}$，求使 $|\sum_{k=1}^{n} \log_{10} f(k)| = 1$ 的所有正整数 n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2015-I-Q09] | Part B (Q6-Q10)

Each ordered starting triple $(a_1, a_2, a_3) \in \{1, \ldots, 10\}^3$ generates $a_n = a_{n-1}|a_{n-2} - a_{n-3}|$ for $n \geq 4$. Count the starting triples for which some term is zero.

每个起始有序三元组 $(a_1, a_2, a_3) \in \{1, \ldots, 10\}^3$ 按 $a_n = a_{n-1}|a_{n-2} - a_{n-3}|$ （$n \geq 4$）生成数列。求最终出现零项的起始三元组数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q10] | Part B (Q6-Q10)

An infinite strictly increasing sequence of positive integers has each triple $(a_{2k-1}, a_{2k}, a_{2k+1})$ geometric and each $(a_{2k}, a_{2k+1}, a_{2k+2})$ arithmetic. Given $a_{13} = 2016$, find a_1.

一个无限严格递增正整数数列中，每个 $(a_{2k-1}, a_{2k}, a_{2k+1})$ 为等比数列，每个 $(a_{2k}, a_{2k+1}, a_{2k+2})$ 为等差数列。已知 $a_{13} = 2016$，求 a_1。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-II-Q09] | Part B (Q6-Q10)

An increasing positive-integer arithmetic sequence a_n and an increasing positive-integer geometric sequence b_n both start at1. Put $c_n = a_n + b_n$. For some integer k, $c_{k-1} = 100, c_{k+1} = 1000$. Find c_k.

递增正整数等差数列 a_n、递增正整数等比数列 b_n 均从1开始，令 $c_n = a_n + b_n$。某整数 k 满足 $c_{k-1} = 100, c_{k+1} = 1000$。求 c_k。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2020-II-Q06] | Part B (Q6-Q10)

Define $t_1 = 20, t_2 = 21$ and $t_n = (5t_{n-1} + 1)/(25t_{n-2})$ for $n \geq 3$. Write $t_{2020} = p/q$ reduced and find $p + q$.

数列满足 $t_1 = 20, t_2 = 21$，且 $n \geq 3$ 时 $t_n = (5t_{n-1} + 1)/(25t_{n-2})$。将 t_{2020} 约分为 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2002-II-Q11] | Part C (Q11-Q15)

Two distinct convergent real geometric series each sum to 1 and have the same second term. One has third term $1/8$. Their common second term is $(\sqrt{m}-n)/p$ with squarefree m. Find $100m+10n+p$.

两个不同的收敛实等比级数和都为 1，且第二项相同，其中一个第三项为 $1/8$。共同第二项为 $(\sqrt{m}-n)/p$，m 无平方因子，求 $100m+10n+p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2005-II-Q11] | Part C (Q11-Q15)

A real sequence has $a_0 = 37, a_1 = 72$, $a_m = 0$, and $a_{k+1} = a_{k-1} - 3/a_k$ for $1 \leq k < m$. Find the positive integer m.

实数列满足 $a_0 = 37, a_1 = 72, a_m = 0$，且 $1 \leq k < m$ 时 $a_{k+1} = a_{k-1} - 3/a_k$，求正整数 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2006-I-Q15] | Part C (Q11-Q15)

A real sequence satisfies $x_0 = 0$ and $|x_k| = |x_{k-1} + 3|$ for $k \geq 1$. Minimize $|\sum_{k=1}^{2006} x_k|$.

实数列满足 $x_0 = 0, |x_k| = |x_{k-1} + 3|$，$k \geq 1$。求 $|\sum_{k=1}^{2006} x_k|$ 的最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2006-II-Q11] | Part C (Q11-Q15)

The tribonacci sequence has $a_1 = a_2 = a_3 = 1$ and $a_{j+3} = a_{j+2} + a_{j+1} + a_j$. Given $a_{28} = 6090307, a_{29} = 11201821, a_{30} = 20603361$, find $\left(\sum_{j=1}^{28} a_j\right) \bmod 1000$.

数列前三项为一，且 $a_{j+3} = a_{j+2} + a_{j+1} + a_j$。已知 $a_{28} = 6090307, a_{29} = 11201821, a_{30} = 20603361$，求前二十八项和除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2007-I-Q14] | Part C (Q11-Q15)

Let $a_1 = a_2 = 3$ and $a_{n+1}a_{n-1} = a_n^2 + 2007$ for $n \geq 2$. Find $\left\lfloor (a_{2007}^2 + a_{2006}^2)/(a_{2007}a_{2006}) \right\rfloor$.

令 $a_1 = a_2 = 3$，$n \geq 2$ 时 $a_{n+1}a_{n-1} = a_n^2 + 2007$。求 $\left\lfloor (a_{2007}^2 + a_{2006}^2)/(a_{2007}a_{2006}) \right\rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2007-II-Q12] | Part C (Q11-Q15)

An increasing geometric sequence consists of integer powers of three, satisfies $\sum_{j=0}^{7} \log_3 x_j = 308$ and $56 \leq \log_3(\sum_{j=0}^{7} x_j) \leq 57$. Find $\log_3 x_{14}$.

递增等比数列各项均为三的整数次幂，满足前八项对数和为 308，且 $56 \leq \log_3(\sum_{j=0}^{7} x_j) \leq 57$。求 $\log_3 x_{14}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q11] | Part C (Q11-Q15)

Let D_n be the determinant of the $n \times n$ tridiagonal matrix with diagonal entries 10, entries immediately above and below the diagonal 3, and all others zero. Reduce $\sum_{n=1}^{\infty}(8D_n+1)^{-1}=p/q$ and find $p+q$.

令 D_n 为 n 阶三对角矩阵的行列式：主对角线均为十，上下紧邻对角线均为三，其余为零。将 $\sum_{n=1}^{\infty}1/(8D_n+1)$ 约为 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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